Cpts 317. 9/19.

HW Hint:

Prob 1. Use structural induction.

$End(L,a)$ is reg if L is.

1. when $L=\emptyset$, $End(L,a)=\emptyset$ is reg;

2. when $L=\{\lambda\}$, $End(L,a)=\emptyset$ is reg,

3. when $L=\{b\}$, $b\in\Sigma$, $End(L,a)=\begin{cases}\{b\} & a=b\\ \emptyset & o.w\end{cases}$ is reg.

4. (I.H.) Assume that L_1, L_2 are reg, and $End(L_1,a)$, $End(L_2,a)$ are reg. Then

4.1, 4.2, 4.3

4.1. $End(L_1+L_2,a)=End(L_1,a)+End(L_2,a)$ is reg. since $End(L_1,a)$ and $End(L_2,a)$ are reg, and reg. is closed under "+".

4.2. $End(L_1 \cdot L_2, a) = ?$

Case 1. $\Lambda \notin L_2$. Then, $End(L_1 L_2, a) = L_1 \cdot End(L_2, a)$ is reg by I.H.;

Case 2. $\Lambda \in L_2$. Then, $End(L_1 L_2, a) = L_1 \cdot End(L_2, a) + End(L_1, a)$ is reg from I.H & reg. is closed under $\cdot$ and $+$.

$L_1 = \{cd, aba\}$, $L_1 L_2 = \{cda, cabb, abaa, ababb\}$

$L_2 = \{a, bb\}$.

$$End(L_1 \cdot L_2, a) = \{cda, abaa\}.$$

$$= L_1 \cdot End(L_2, a)$$

However.

$$End(L_1 \cdot L_2, a) = \{cda, abaa\} \neq$$

$$L_1 \cdot End(L_2, a) + End(L_1, a)$$

$$= \{cda, abaa\} + \{aba\}$$

4.3. $\mathrm{End}(L_1^*, a) = L_1^* \cdot \mathrm{End}(L_1, a)$

is reg for $I \setminus H$ & reg is closed under $*$ and $\cdot$.

Prob 2. $L = ((aa+bbb)^* c)^*$

$Ed(L, a) = \emptyset$.

Prob 3. do it. structural induction.

Prob 4. I give you another example:

$L = (abc + ab)^*$

$L^r = (cba + ba)^*$.

$L = ab(abc+ab)^*$

$L^r = (cba+ba)^* ba$

Prob 5. I try another exmple:

tell me shortest words in

$$(ab + ba)^* ab.$$

$$\|$$

$$\wedge$$

tell me shortes words in

$$(ab^* + b^*a^*)^* (ab + a)$$

$$\wedge \quad a$$

Prob 6. Given: r as a regExp

Question: all shortest words in r.

Alg:

1. if $r = \emptyset$, ret $\emptyset$;
2. if $r = \Lambda$, ret Λ;
3. if $r = a$, $a \in \Sigma$, ret a;
4. Let Shortest(L_1) and Shortest(L_2) are shortest words in L_1 & in L_2 resp, and L_1 & L_2 are reg;

4.1 Shortest($L_1 + L_2$) = shortest in shortest(L_1) ~~and~~ OR shortest(L_2);

4.2 Shortest($L_1 L_2$) = Shortest(L_1) · Shortest(L_2);

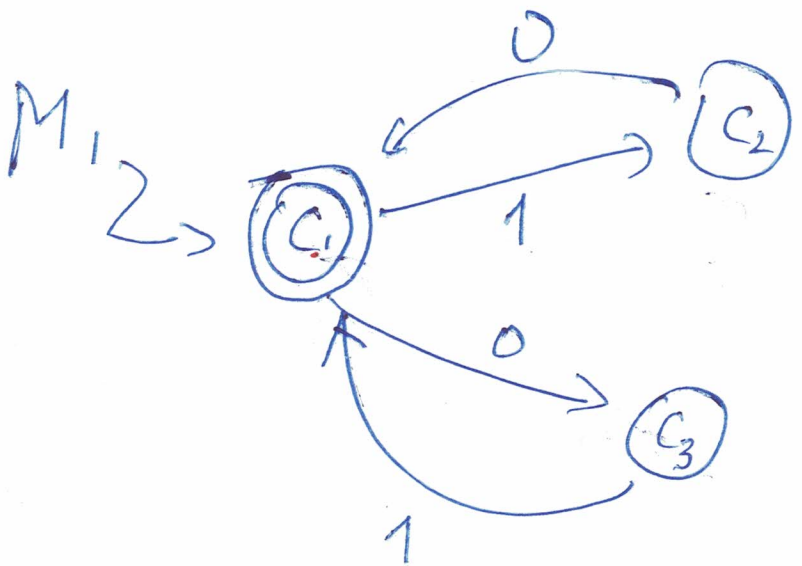
4.3. Shortest(L_1^*) = Λ;

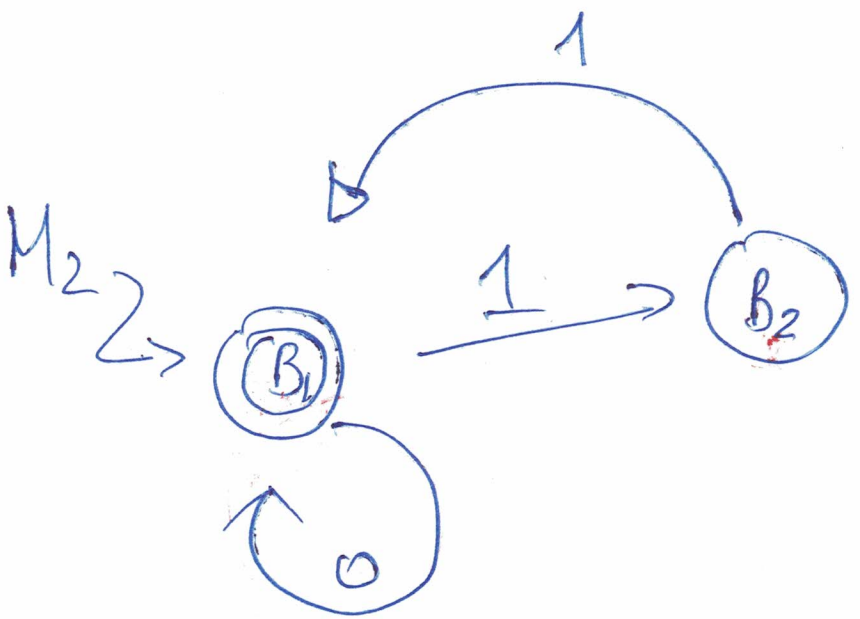
Last time: Given: DFA M_1 & M_2

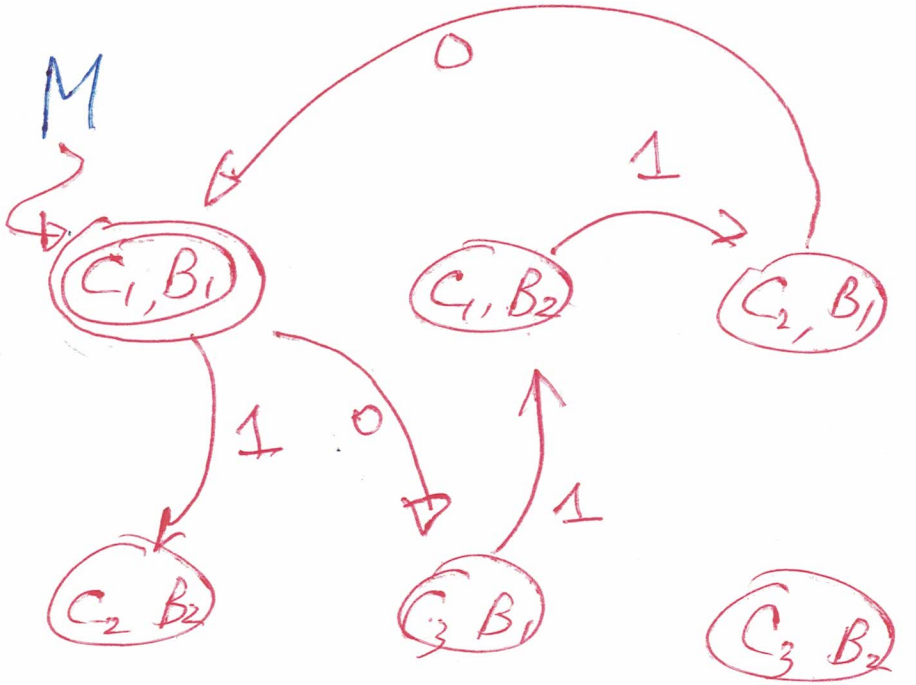
we can construct DFA M s.t.

$$L(M) = L(M_1) \cap L(M_2).$$

Cartesian Product construction: (Exaple).







$L(M) = (0110)^*$.

$L(M_1) = (10+01)^*$

$L(M_2) = (0+11)^*$.

So,

$(10+01)^* \cap (0+11)^*$

$= (0110)^*$.